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# PALEOMAGNETIC EVIDENCE FOR EN ECHELON CRUSTAL EXTENSION AND CRUSTAL ROTATIONS IN WESTERN MONTANA AND IDAHO

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Abstract. The Bitterroot metamorphic core complex is one of many Eocene core complexes in the northern Cordillera. A middle Eocene dike swarm cuts the Bitterroot metamorphic core complex and its hanging wall, the Skalkaho slab. A 26° change in strike of the dikes across the core complex suggests either a refracted stress field or rotation during Eocene or younger extension. The nearby Bitterroot Valley and Big Hole Basin also indicate tectonic rotation. To distinguish between the hypotheses of refracted stress and tectonic rotation, we collected paleomagnetic data from 28 dikes. Nineteen sites provide reliable results. Eight sites from the hanging wall east of the mylonite zone give Dec = 46°, Inc = 69°, $\alpha_{95}$ = 13°; 11 dikes to the west yield Dec = 334°, Inc = 64°, $\alpha_{95}$ = 6°. The declinations diverge from the expected direction; the inclinations do not. The hanging wall rocks are rotated 58° ± 32° clockwise at the 95% confidence level. The footwall shows 14° ± 11° of counterclockwise rotation. Both normal and reversed polarity dikes are present in each area, indicating that the dike swarms may adequately average paleosecular variation. Vertical contacts reveal that the dikes have not tilted since emplacement. These paleomagnetic results and available geochronologic data show that the Bitterroot dome was pulled from beneath the Skalkaho slab in the Eocene. As the dome slid toward the northwest, it experienced a slight counterclockwise rotation, while the upper plate moved clockwise around a vertical axis near its northern edge. Rotation of the Skalkaho slab occurs because the slab lies between en echelon extensional detachments of the core complex and the next basin to the east, the Big Hole Basin. The Boehls Butte area, north central Idaho, occupies a step in the Lewis and Clark fault zone and is probably a core complex related to right-lateral motion on the fault system. Similarities in both the style and timing of extension suggest that Eocene ductile/brittle extension occupies a swath from British Columbia, past the Bitterroot metamorphic core complex, to the Pioneer core complex in south central Idaho.

## INTRODUCTION

The Bitterroot metamorphic core complex and Tertiary basins of western Montana are part of an Eocene extensional system stretching from British Columbia to Idaho (Figure 1a).

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The age of the Bitterroot complex is early to middle Eocene [Bickford et al., 1981; Chase et al., 1983; Garmezy, 1983]. Chase and Talbot [1973], Hyndman et al. [1975], and Hyndman [1980] divided the core complex into three parts: (1) the Bitterroot dome, (2) the Bitterroot mylonite, and (3) the upper plate (Figure 1b). Hyndman et al. [1975] referred to the upper plate as the Sapphire tectonic block, a Late Cretaceous feature. We refer to a similar crustal domain, of greater regional extent and Tertiary in age, as the Skalkaho slab.

The Bitterroot mylonite's kinematics show that the Skalkaho slab/Sapphire tectonic block once overlay the Bitterroot dome. However, the timing and tectonics of the separation are equivocal. Hyndman et al. [1975] and Hyndman [1980] proposed that the Sapphire tectonic block slid to the east during the Late Cretaceous. Yet, Chase et al. [1983] and Garmezy [1983] suggest that the Bitterroot dome was pulled westward during the Eocene.

A dike swarm of middle Eocene age (Figure 1b) intrudes the Bitterroot dome and the Skalkaho slab. The average strike of the dike swarm is N21°E in the Bitterroot dome and N47°E in the Skalkaho slab. The divergent strikes imply that either (1) the Bitterroot dome and/or the Skalkaho slab moved since emplacement or (2) the dikes intruded in a refracted stress system. To test these hypotheses, we conducted a paleomagnetic study of the dikes. Our results indicate the Skalkaho slab rotated after emplacement of the dikes.

## REGIONAL GEOLOGY

The Bitterroot metamorphic core complex is the northernmost metamorphic core complex associated with the Basin and Range province. The core complex developed along the eastern flank of the Idaho batholith as the northern part of the Idaho batholith and associated country rocks moved to the northwest during Tertiary extension [Hamilton and Myers, 1966]. Left behind was thinned continental crust [Sheriff and Stickney, 1984] diced with extensional basins up to 5 km deep. The Bitterroot mylonite, Lewis and Clark fault zone, and Tertiary basins all participated in the crustal extension. Granites, rhyolites, and minor andesites were emplaced across western Montana and Idaho at about the same time.

### Bitterroot Metamorphic Core Complex and the Skalkaho Slab

The Bitterroot mylonite [Chase and Talbot, 1973] is the dominate structure in the Bitterroot metamorphic core complex. The east dipping mylonite separates deformed, highly metamorphosed rocks beneath the mylonite from low-grade rocks in the hanging wall. The mylonite has a basal amphibolite-facies section that passes discordantly upward into the greenschist facies. Chloritic breccia caps the mylonites [Hyndman and Meyers, 1988]. All three facies display top-to-the-east (110°) shear [Garmezy, 1983; Hyndman and Meyers, 1988].

The Bitterroot dome, on the northeastern flank of the Idaho batholith, is the metamorphic core beneath the mylonite. Mesozonal granodiorites and granites of the Idaho batholith are prevalent in the core; they range in age from 81 to 46.5 Ma [Bickford et al., 1981; Ferguson, 1975]. Country rocks are in the upper amphibolite facies and may include crystalline basement rocks. An arcuate line of middle Eocene (51 Ma)

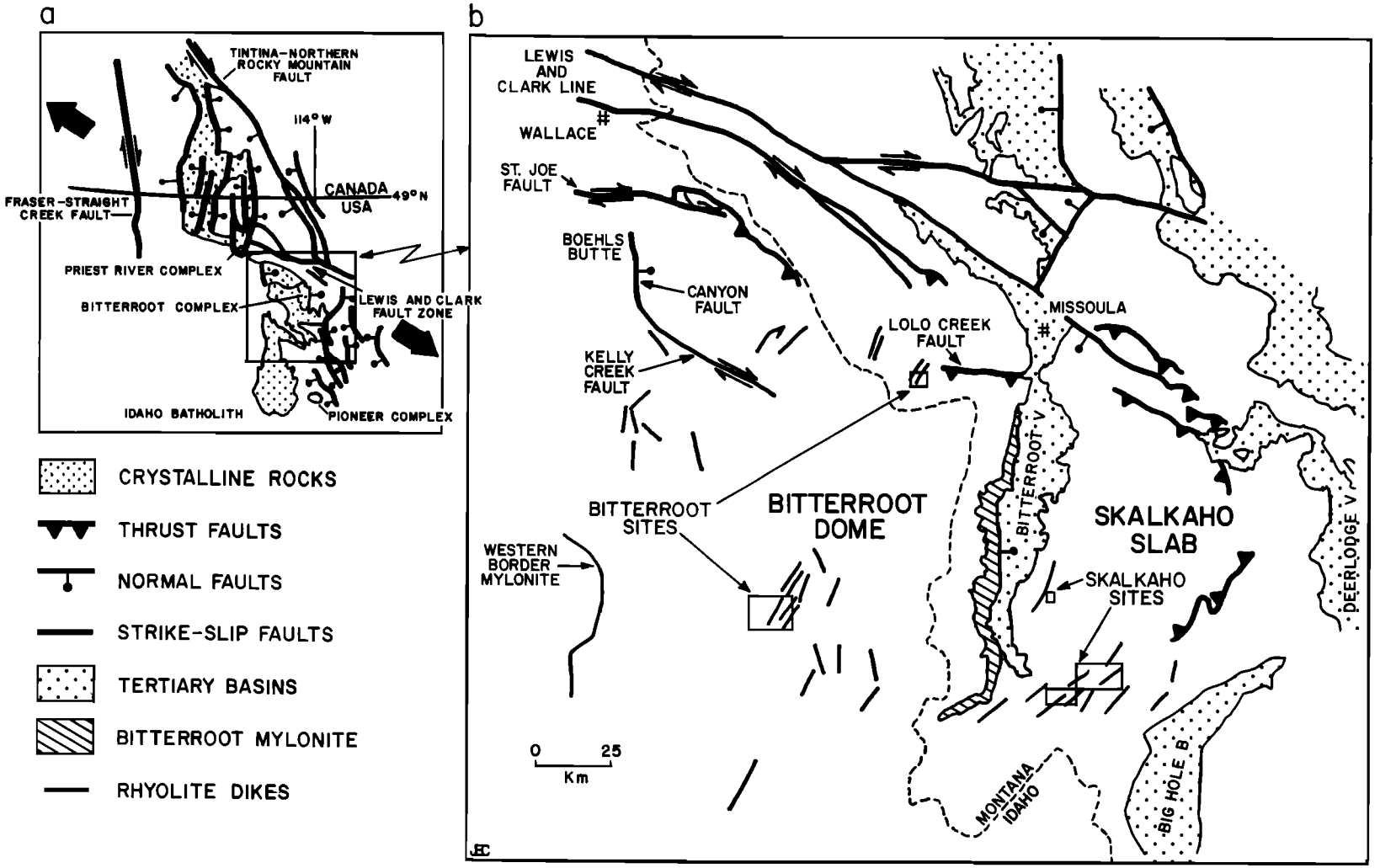



Fig. 1. Simplified geologic map of (a) the northern Cordillera (a) and (b) an enlargement for western Montana and Idaho. In Figure 1b, the Eocene dike swarm is enlarged to about 4 times actual size for clarity. See Doughty [1990] for the sources used in map compilation.

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epizonal granites and associated rhyolite dikes crudely define the western edge of the dome [Lund, 1980].

Hyndman et al. [1975] named the structurally higher region east of the mylonite the Sapphire tectonic block. As defined, the Sapphire tectonic block is a supracrustal allochthon bounded by Late Cretaceous faults. However, new radiometric dates, regional relationships, and our paleomagnetic data show that faults with significant Tertiary displacement bound the rocks above the Bitterroot mylonite. We wish to avoid the confusion caused by a new definition for an established name. Hence we call the Tertiary tectonic block the Skalkaho slab; Figure 1b shows its boundaries.

Some argue that the amphibolite-facies mylonite is Late Cretaceous, others say Eocene. To resolve this, Bickford et al. [1981] and Chase et al. [1983] obtained U-Pb zircon and monazite ages of synkinematic granodiorites and pegmatites [Latour and Barnett, 1987] from the shear zone. Lower intercept ages from these rocks are 48 ± 1 Ma to 57 ± 4 Ma [Chase et al., 1983; Toth, 1983]. Most are discordant lower intercept ages from zircons with detrital cores and euhedral overgrowths [Bickford et al., 1981]. One sheared pegmatite yields a concordant U-Pb zircon age of 52 ± 1 Ma [Chase et al., 1983]. The ages from zircons, monazites, and field relationships are all consistent. This argues for a faithful representation of the crystallization age [Bickford et al., 1981]. Also, $^{40}Ar/^{39}Ar$ dates from hornblende and biotite in the mylonite zone reveal rapid cooling and uplift with mylonitization from 43.5 to 45.5 Ma [Garmezy, 1983]. These data demonstrate that the amphibolite-facies mylonite was active during at least the early to middle Eocene. The greenschist-facies mylonite and chloritic breccia cannot be any older.

The Bitterroot dome, like other core complexes [Parrish et al., 1988], experienced thermal disruption during the Eocene. In the northern Bitterroot dome, K-Ar and $^{40}Ar/^{39}Ar$ dates from biotite cluster at 40 to 44 Ma [Armstrong et al., 1977; Williams, 1979; Ferguson, 1975; Garmezy, 1983; Garmezy and Sutter, 1983; Hayden and Wehrenberg, 1960]. To the south, $^{40}Ar/^{39}Ar$ dates show that hornblende cooled through its blocking temperature at 51 ± 5 Ma [Garmezy, 1983]. Biotite cooling ages from the same rocks are much younger at 42 to 45 Ma [Garmezy, 1983]. In contrast, the Skalkaho slab has Late Cretaceous and Paleocene cooling ages [McDowell and Kulp, 1969; Hyndman et al., 1972]. Thus the cooling history of the Bitterroot dome reveals rapid uplift during Eocene mylonitization, while the Skalkaho slab's cooling ages do not.

### Lewis and Clark Fault Zone

The Lewis and Clark fault zone was an important element during the Tertiary extension of western Montana. This system of strike-slip faults extends from the Priest River complex, in eastern Washington, to central Montana, where strike-slip motion passes into several north trending extensional basins [Reynolds, 1979] (Figure 1a). En route, the swarm passes north of the Bitterroot dome and forms the northern edge of the Skalkaho slab. Before the Tertiary, the Lewis and Clark fault zone was a left-lateral strike slip system [Hyndman et al., 1988]. Tertiary movement on the fault zone is right lateral; the cumulative displacement is between 35 and 81 km [Harrison et al., 1974; Sheriff et al., 1984]. The fault system was active throughout the development of the Bitterroot metamorphic core complex.

### Epizonal Eocene Magmatism and the Eocene Dikes

A swarm of rhyolitic, hypabyssal dikes is part of middle Eocene (51-44 Ma) plutonism and volcanism in Montana and Idaho [Armstrong, 1974; Armstrong et al., 1977; Hyndman et al., 1977]. The dikes are apophyses of shallow granites and fed overlying volcanic rocks [Bennett, 1980; Badley, 1978]. The physical relationship between the dikes and the Bitterroot mylonite is not well understood [cf. Holloway, 1980]. Yet the geochronologic data show that these rocks are coeval with the synkinematic granodiorites and pegmatites injected into the Bitterroot mylonite zone. Thus epizonal Eocene magmatism also accompanied motion on the Bitterroot mylonite.

In the Skalkaho slab, the dike swarm strikes N47°E. The dikes and an Eocene pluton are well aligned. In the Bitterroot dome the dikes strike N21°E; this average strike has more variance than that for the Skalkaho slab. The western swarm is perpendicular to the 110° stretching lineation in the Bitterroot mylonite and probably formed in the same stress field. The 26° difference in the strike of the Eocene dike swarms occurs across the Bitterroot mylonite. The dikes converge toward the south (Figure 1b); the Bitterroot Valley also tapers to the south. These relations suggest that some separation of the Bitterroot dome and Skalkaho slab is post-middle Eocene in age. Such divergence is consistent with tectonic rotation about a nearby vertical axis during displacement on the mylonite.

## PALEOMAGNETIC FIELD AND LABORATORY WORK

To test the hypothesis of differentially rotated dikes, we collected samples from 28 paleomagnetic sites. Fourteen sites, from two sampling areas, are in the Bitterroot dome. The remaining 14 sites are from two areas in the Skalkaho slab (Figure 1b).

At 20 sites we drilled samples with a portable rock drill and oriented them in situ with magnetic and sun compasses. Most of these sites, sampled perpendicular to strike, are in roadcuts. However, eight sites in the Bitterroot dome are from the Selway-Bitterroot Wilderness area. At these sites, we collected one or two hand samples from each dike.

For each site, we subjected one pilot specimen to stepwise thermal demagnetization and one to alternating field (af) demagnetization. Usually, thermal demagnetization isolated the same direction as af demagnetization. A Molspin tumbling demagnetizer performed the af demagnetization. Thermal demagnetization was completed with a homemade device; induction in the cooling chamber is less than 6 nT. We used the technique that best isolated the characteristic remanent direction to progressively demagnetize the remainder of the site. For each specimen, 10-25 steps were measured with a Schonstedt SSM-2A magnetometer.

Orthogonal projections simplified the interpretation of each specimen's progressive demagnetization. Most specimens had single-component characteristic remanent directions. Only 13 of 170 specimens contained more than one persistent magnetic component. We used intersecting remagnetization circles [Halls, 1978] to isolate the common component of magnetization in these specimens. Figure 2 shows representative demagnetization diagrams for our reliable specimens.

Site mean directions were calculated using the methods of Fisher [1953], Kirschvink [1980], and Onstott [1980]. For hand-sampled sites, we combined directions from each

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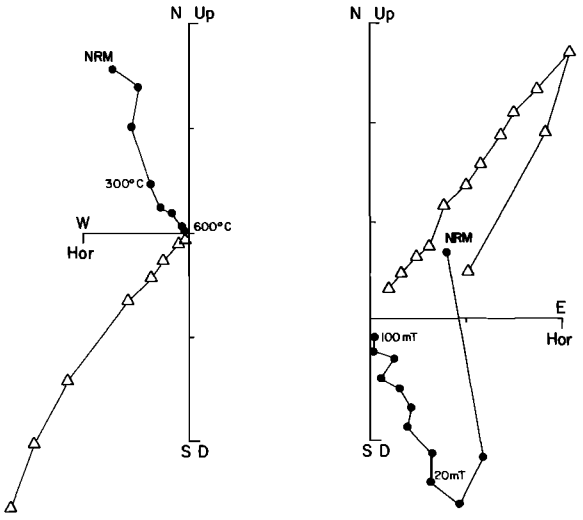



Fig. 2. Vector endpoint diagrams of typical specimens during demagnetization. Circles (triangles) represent declination (inclination). Open (closed) symbols represent upper (lower) hemisphere. One scale division equals $10^{-5}$ emu/cc.

specimen in a hand sample to get a mean direction. Averaging these mean directions from hand samples produced site mean directions.

To get site mean directions that accurately reflect the original magnetic field, it is common to apply arbitrary rejection criteria to specimens and sites. Our specimens had to be within two angular standard deviations of the site mean direction. They also had to have maximum angular deviation values less than 5° or 15° for linear and arcuate decay paths, respectively. These criteria eliminated 40 specimens, 130 survived.

We also dropped nine of 28 sites from further statistical analysis. Specimens from five sites were too weak ($M_0 < 10^{-4}$ A/m) to measure with our equipment. Two sites record both normal and reverse polarities in transects across the dike. One of these has inclinations ranging from 60° to -10° across the dike. These dikes probably gained their magnetization over enough time to record some anomalous field behavior. By analogy, most of our dikes likely averaged some secular variation. The sites we drilled in the field have 95% confidence limits ($\alpha_{95}$) less than 15°. However, our hand-sampled sites have few independent blocks. Thus $\alpha_{95}$ is not an appropriate rejection criterion. We used an angular standard deviation ($\delta_{63}$) of 15° as the arbitrary cutoff for hand-sampled sites. Our reliable drilled sites also had $\delta_{63}$ below 15°. This cost us two hand-sampled sites. Table 1 presents the results from each site.

## PALEOMAGNETIC RESULTS

Most samples lose about 60% of their magnetization by 300°C or 50 mT, the rest disappears near 580°C. This magnetization resides in magnetite. However, for 27 of 79 thermally cleaned specimens about 15% of the natural remanent magnetization (NRM) persists until 680°C, the unblocking temperature of hematite. The direction of magnetization carried by hematite usually parallels that in the magnetite. Such hematite likely grew during deuteric alteration. Rapid acquisition of isothermal remanent magnetization (IRM) [Doughty, 1990] shows that magnetites are the main magnetic carrier.

A weak secondary component of magnetization exists in 56 specimens from 12 sites. For seven of these sites the component is a viscous remanent magnetization (VRM) parallel to the present geomagnetic field. For the remaining five sites, the secondary components have more dispersion. Magnetic cleaning removes the secondary components by 300°C or 50 mT.

Doughty [1990] presents the details of some rock magnetic experiments that lead to the following results. High NRM:IRM(s) ratios from our specimens show the magnetization is of primary origin [Fuller et al., 1988]. Many of the dikes have large, possibly multidomain, grains. However, Lowrie and Fuller [1971] domain-state tests show that the specimens also have single-domain grains. These grains probably carry the characteristic remanent magnetizations in our specimens. Wu et al. [1974] and Murthy et al. [1971] investigated similar situations. Their results also show that single-domain and pseudo single-domain magnetite grains exist in coarse-grained igneous rocks.

Determining paleohorizontal is difficult with igneous rocks, yet it is necessary for accurate reconstructions. We make the assumption that dikes intrude vertically. Our support for this assumption is based on the fact that the dike swarms outcrop linearly over tens of kilometers along strike and most dikes have vertical contacts. However, some dikes have inclined contacts. Where these dikes are well exposed, they are rolling over to feed contiguous sills. Thus their dip is not a postemplacement feature. On the basis of these criteria, we infer that tectonic tilt of the dike swarms is unlikely.

A modified tilt test [McFadden and Lowes, 1981] on seven inclined dikes also shows that the tilt is original. Restoring tilted dikes from the Skalkaho slab to vertical is inconclusive. The dispersion of site mean directions increases, but only at the 80% confidence level. However, for tilted dikes in the Bitterroot dome, the increase in dispersion is significant at the 95% confidence level. The tilted dikes were probably magnetized in situ. Thus the paleomagnetic results in Table 1 do not require or include tilt corrections.

To apply paleomagnetic results to tectonic problems, paleosecular variation must be averaged to a constant value. Paleosecular variation model G [McFadden et al., 1988] predicts 17.5° of angular standard deviation ($\delta_{63}$) from virtual geomagnetic poles (VGP) at our sampling paleolatitude. Because the dikes are as much as 20 m thick and cooled slowly, each averages some paleosecular variation. Thus between site scatter of VGP should be less than predicted by model G. The Bitterroot area yields $\delta_{63} = 12.8°$ with 95% confidence limits of -9.3° and +19.9° [Cox, 1969]. Sites in the Skalkaho slab give $\delta_{63} = 27.0°$ (-20.4°, +40.0°). Neither are directly on the expected value. For two additional reasons, we believe the Bitterroot and Skalkaho localities may adequately average paleosecular variation. First, normal and reversed polarity dikes are present in both areas. In the Skalkaho slab, three dikes have normal polarity and five reversed polarity. The Bitterroot dome has eight normal polarity dikes and three of reverse polarity. Thus the sites represent a time span greater than the longest period of secular variation. Second, as we discuss below, some of our declinations are anomalous, yet the inclinations are concordant. If the unusual declinations result from incomplete

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TABLE 1. Site Mean Paleomagnetic Data, Bitterroot Dome/Skalkaho Slab

| Site | Lat | Long | N/Nc | Inc | Dec | α95 | δ63 | k | Plat | Plong |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Bitterroot Localities | | | | | |
| B01 | 46.7 | 114.6 | 4/6 | 65.7 | 336.9 | 9.5 | 8.3 | 94.5 | 74.2 | 169.6 |
| B02 | 46.7 | 114.6 | 4/8 | 66.9 | 342.8 | 11.9 | 10.4 | 60.4 | 78.3 | 175.6 |
| B03 | 46.7 | 114.6 | 7/8 | 59.0 | 324.8 | 3.8 | 5.0 | 256.8 | 63.8 | 153.3 |
| B05 | 46.7 | 114.6 | 5/6 | 59.2 | 331.4 | 11.1 | 11.6 | 48.6 | 69.4 | 150.2 |
| B06 | 46.7 | 114.6 | 3/5 | 50.9 | 327.1 | 10.3 | 6.7 | 144.3 | 60.7 | 136.3 |
| B09 | 46.1 | 115.2 | 2/2# | -64.6 | 171.7 | 35.9 | 11.4 | 50.5 | -84.2 | -17.6 |
| A | | | 2/4 | -72.5 | 166.5 | 19.1 | 6.2 | 172.9 | -75.7 | 34.6 |
| B | | | 2/3 | -56.7 | 174.5 | 16.5 | 5.3 | 231.4 | -80.3 | -88.2 |
| B10 | 46.1 | 115.2 | 3/4 | 71.4 | 344.0 | 17.9 | 11.6 | 48.5 | 75.9 | -154.3 |
| B11 | 46.1 | 115.1 | 2/2# | 78.1 | 328.1 | 47.3 | 14.8 | 30.0 | 63.2 | -142.2 |
| A | | | 4/5 | 81.8 | 268.3 | 13.3* | 10.9* | -31/-.9 | 43.3 | -137.5 |
| B | | | 4/5 | 69.2 | 348.4 | 11.0 | 7.2 | 126.0 | 80.0 | -159.3 |
| B12 | 46.1 | 115.1 | 2/2# | -57.7 | 163.0 | 6.8 | 2.2 | 1362.9 | -75.5 | -50.8 |
| A | | | 5/5 | -57.7 | 160.1 | 4.2 | 4.4 | 333.0 | -73.4 | -45.7 |
| B | | | 3/3 | -57.6 | 165.9 | 2.0 | 1.3 | 3905.3 | -77.0 | -56.9 |
| B15 | 46.1 | 115.1 | 2/2# | -66.9 | 120.3 | 40.4 | 12.7 | 40.4 | -50.9 | 2.5 |
| A | | | 5/5 | -64.3 | 99.3 | 6.9* | 6.3* | -120/-.3 | -36.6 | -56.5 |
| B | | | 2/3 | -66.4 | 143.2 | 3.8 | 1.2 | 4447.8 | -65.2 | -5.2 |
| B18 | 46.1 | 115.1 | 8/8 | -54.8 | 161.8 | 2.3 | 3.3 | 599.5 | -72.6 | -56.8 |
| | | | | | Skalkaho Localities | | | | | |
| S03 | 45.8 | 113.9 | 7/8 | -71.9 | 270.1 | 5.7 | 7.6 | 113.4 | -36.8 | 108.3 |
| S04 | 45.9 | 113.9 | 7/8 | -60.3 | 233.7 | 6.1 | 8.1 | 99.7 | -51.5 | 142.8 |
| S07 | 45.9 | 113.8 | 6/8 | 67.8 | 350.9 | 5.7 | 6.9 | 138.0 | 82.2 | -161.5 |
| S10 | 45.9 | 113.7 | 5/8 | 75.0 | 100.5 | 6.4 | 6.8 | 142.8 | 35.3 | -79.8 |
| S12 | 45.9 | 113.8 | 6/8 | 79.1 | 90.6 | 4.3 | 5.1 | 247.3 | 41.8 | -84.8 |
| S13 | 45.9 | 113.8 | 5/7 | -54.5 | 197.2 | 11.0 | 11.5 | 49.5 | -73.3 | -170.6 |
| S16 | 46.1 | 114.0 | 3/7 | -64.1 | 186.2 | 14.0 | 9.2 | 78.2 | -86.1 | 154.8 |
| S17 | 46.1 | 114.0 | 6/8 | -44.9 | 247.8 | 2.8 | 3.3 | 596.3 | -33.8 | -19.5 |

A and B denote hand samples from hand-sampled sites (only one hand sample was used at sites B18 and B10); Lat and Long are sampling latitude and longitude, positive north and west, respectively; N/Nc is number of specimens averaged/collected (# denotes number of hand samples averaged/collected); Inc. and Dec. are inclination and declination of site mean direction in degrees; α95 is radius of 95% confidence cone in degrees; δ63 is circular standard deviation in degrees; k is Fisher [1953] precision parameter; Plat. and Plong. are VGP latitude and longitude. The asterisk denotes average of major and minor axes of α95 ellipse and δ63 oval from analysis of remagnetization circles, concentration parameters [Onstott, 1980] for such data are tabulated as $k_1/k_2$.

TABLE 2. Locality Mean Paleomagnetic Results From the Bitterroot Dome and Skalkaho Slab, Idaho and Montana

| Locality | N | $\propto_{95}$ | k | Dec | Inc | R | ± R | F | ± F | |
|---|---|---|---|---|---|---|---|---|---|---|
| Bitterroot | 11 | | 5.6 | 68.1 | 334.1 | 63.8 | -14.4 | 11.0 | 1.7 | 5.2 |
| Skalkaho | 8 | | 13.4 | 18.1 | 46.4 | 69.1 | 57.9 | 32.0 | -3.9 | 10.7 |

N is number of sites; k is Fisher [1953] precision parameter; $\propto_{95}$ is radius of 95% confidence cone in degrees; Inc. and Dec. are inclination and declination of locality mean direction in degrees; $R = D_{ob} - D_{ex}$; ± R = radius of 95% confidence cone; $F = I_{ex} - I_{ob}$; ± F = radius of 95% confidence cone. Expected declination ($D_{ex}$) = 348.5°; expected inclination ($I_{ex}$) = 65.5° [Diehl et al., 1983].

averaging of paleosecular variation, we expect discordant inclinations as well.

Table 2 presents mean directions for the Bitterroot dome and Skalkaho slab. For the Bitterroot dome, the mean direction from 11 sites is Dec = 334°, Inc = 64°, $\alpha_{95}$ = 6°. Eight sites from the Skalkaho slab give Dec = 46°, Inc = 69°, $\alpha_{95}$ = 13° (Figure 3).

## TECTONIC IMPLICATIONS

### The Bitterroot Metamorphic Core Complex

Diehl et al. [1983] provide a North American reference pole (at 82.0°N and 170.2°E, $\alpha_{95}$ = 3.5°) for the period from 47 to 54 Ma. Using this paleopole, the expected

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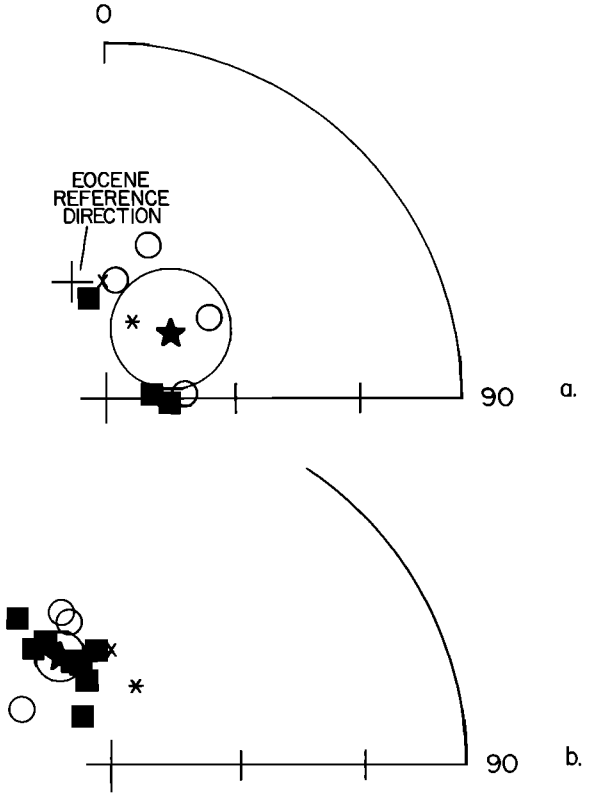



Fig. 3. Lower hemisphere equal-area projections of the (a) Skalkaho and (b) Bitterroot locality site mean directions. All reversed polarity site mean directions have been inverted into the lower hemisphere and portrayed as open circles; solid squares represent normal polarity site mean directions; Pentagram and large circle are locality mean direction and 95% confidence cone in degrees; plus sign is the expected Eocene direction from Diehl et al. [1983]; small cross and asterisk are the current dipole and local geomagnetic field directions, respectively.

paleomagnetic direction in our study area is Dec = 348.5°, Inc = 65.5°. When compared to the reference direction through standard techniques [Beck, 1980; Demarest, 1983], the sites in the Bitterroot dome show 14° ± 11° of counterclockwise rotation at the 95% confidence level. Flattening of inclination for the Bitterroot region is 2° ± 5° (positive is shallower). The Skalkaho slab shows 58° ± 32° of clockwise rotation and -4° ± 11° of flattening. Perhaps our sites in the Skalkaho slab do not completely average paleosecular variation. However, the concordant inclinations make it unlikely that all the difference in declination is from incomplete averaging of paleosecular variation.

If the dikes swarms intruded with a 26° difference in strike, their paleomagnetic directions would be the same. They are not. The divergence of the dikes is most likely the result of tectonic rotation during post-middle Eocene crustal extension. Thus much of the development of the Bitterroot core complex is post-middle Eocene in age.

The Bitterroot dome shows 3° of significant counterclockwise rotation at the 95% confidence level but no flattening. The anomalous declination could be from the footwall beneath the mylonite tilting around a north striking horizontal axis. However, the regional plunge of the Bitterroot dome is to the south, perpendicular to the needed direction. Thus the counterclockwise deflection of declination is probably not from tilting. Most likely the deflected declination measures rotation about a nearby vertical axis. Regardless of rotation, regional geology shows the Bitterroot dome moved to the northwest along the Lewis and Clark fault zone [Hamilton and Meyers, 1966]. For example, Sixt [1988] estimates that the Bitterroot mylonite records at least 25 km of displacement. Also Harrison et al. [1974] propose 81 km of right-lateral offset on the Lewis and Clark fault zone. This is the upper limit of extension in the area. Because the Lewis and Clark fault zone parallels a line of latitude around the Eocene pole, our data do not measure the known translation of the dome. We suggest that as the dome slid northwest some 25 km along the fault zone, it rotated counterclockwise.

The Skalkaho slab shows 58° ± 32° of clockwise rotation. To produce the measured declination with tilting, the Skalkaho slab would have to dip 18° to the west. However, Cretaceous metamorphism within the Skalkaho slab decreases toward the northeast, suggesting that the slab plunges northeast. Again, the regional plunge is perpendicular to the needed direction. Additionally, the dikes outcrop linearly, with vertical contacts, over tens of kilometers of rugged topography. There is little or no tilt to the west. The simplest interpretation is that the Skalkaho slab rotated clockwise about a nearby vertical axis. The Skalkaho slab probably rotated as a coherent slab, bounded by discrete extensional faults.

Structural relationships constrain the location of the rotation axis for the Skalkaho slab. Suppose the rotation axis lay near the south end of the Skalkaho slab. Then, there would be about 44 km of middle Eocene, left-lateral, strike-slip motion along the northern edge of the Skalkaho slab. There is no such motion documented to date. Tertiary displacement on the Lewis and Clark fault zone is on down-to-the-southwest normal faults [Nelson and Dobell, 1961; Desormier, 1975]. These Tertiary normal faults only slightly offset Cretaceous structures and reveal little strike-slip motion. The rotation axis for the Skalkaho slab cannot be at its south end; it must be toward the northern end.

## En Echelon Extension and Tertiary Basins

The Bitterroot mylonite and Tertiary basins also display evidence for crustal rotation. At its northern end, the Bitterroot mylonite separates upper amphibolite-facies rocks in the footwall from low-grade rocks in the hanging wall. Within 100 km to the south, the mylonite juxtaposes rocks of similar metamorphic grade. At least 25 km of displacement wanes within 100 km. This, and rotation of the dome, causes the similar southward narrowing of the Bitterroot Valley.

Counterclockwise rotation of the Bitterroot dome can account for 10 km (40%) of differential displacement on the mylonite. At the north end of the mylonite, there remains 15 km of regional extension along the mylonite. At the south end, all the regional extension (at least 25 km) must be accounted for on other structures.

The most likely candidate to take up regional extension is the Big Hole Basin to the southeast (Figure 1b). This basin is a half-graben that may contain an east dipping Eocene detachment along its western edge. The Big Hole Basin narrows toward the north, opposite from the taper of the

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Bitterroot Valley. Such tapering reflects the fact that extension in the Bitterroot Valley decreases southward, whereas extension in the Big Hole Valley decreases toward the north (Figure 4). Considering both basins, extension is constant across the region. The Big Hole and Bitterroot valleys result from en echelon faults that lack a connecting transform fault.

Accepting the en echelon relationship between the Bitterroot Valley and Big Hole Basin explains the clockwise rotation of the Skalkaho slab. As the Bitterroot dome moved from beneath the Skalkaho slab the Big Hole and Deerlodge basins formed to the southeast. The Skalkaho slab, caught between these basins and the dome, rotated clockwise about a vertical axis at its northern end (Figure 4).

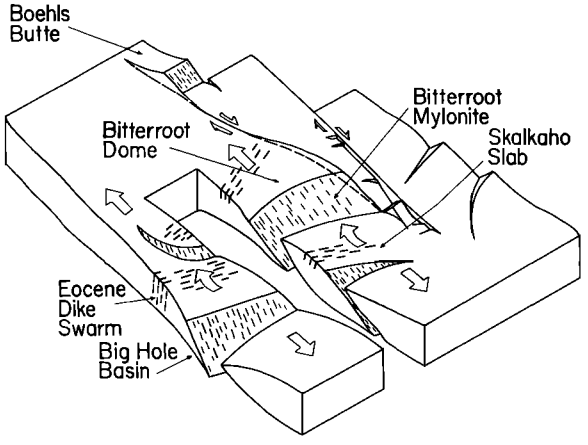



Fig. 4. Block diagram illustrating the proposed scenario for Tertiary extension in western Montana and Idaho. Crust south of the Lewis and Clark line and west of the extensional detachments (lined pattern) has moved to the west by 25-35 km. The Skalkaho slab rotated clockwise about an axis toward the north and the Bitterroot dome rotated slightly counterclockwise as the crust extended along the detachments. Boehls Butte, northwest of the Bitterroot dome, occupies a right step in the Lewis and Clark line and may be a small Eocene core complex.

## Regional Perspective

Rotating the Skalkaho slab around a northern axis requires swinging the southern end toward the west. Yet, no major strike-slip faults bound the southern edge. Instead, regional crustal extension accommodates the displacement. The Trans-Challis fault zone, a northeast trending system of Eocene normal faults, crosses the region [O'Neill and Lopez, 1985]. Eocene ductile extension, like that in the Pioneer core complex [Wust, 1986; O'Neill and Pavlis, 1988], is exposed in fault-bounded ranges in south central Idaho (Figure 1a). Like the structures associated with the Bitterroot core complex, concurrent basins began to form during the middle Eocene [Janecke and Snee, 1990]. Some of the uplifted ranges exhibit paleomagnetic evidence of post-middle Eocene crustal rotations as well [Janecke et al., 1991]. The displacement at the south end of the Skalkaho slab is distributed among these extensional structures.

Northwest of the Bitterroot core complex is a wide zone of Eocene (58-45 Ma) crustal extension in British Columbia and northeast Washington [Parrish et al., 1988; Bickford et al., 1985]. Crustal extension and clockwise rotation occur between two en echelon, right-lateral, strike-slip faults that strike northwest [Price, 1979] (Figure 1a). Our synthesis of the regional geology is that similar crustal rotation and extension occurs east of the Bitterroot core complex and south into east central Idaho. This rotation and extension developed in response to movement of the Idaho batholith extensional terrane toward the northwest [Hamilton and Meyers, 1966] at the same time (57-44 Ma) as extension to the north. The two areas of extension are coupled, in echelon, by the right-lateral Lewis and Clark line which behaved as a continental transform fault during extension [Hamilton and Myers, 1966; Ewing, 1980; Sheriff et al., 1984; Rehrig et al., 1987].

Between the Priest River complex and the Bitterroot metamorphic core complex is Boehls Butte (Figure 1b). Boehls Butte is an enigmatic region of highly metamorphosed rocks with Eocene K-Ar cooling ages [Armstrong et al., 1977; Hietanen, 1984]. Boehls Butte occupies a right-stepping jog (Figures 1b and 4) in the Lewis and Clark fault zone, and crustal extension is expected in such locations. We suspect the Boehls Butte area is an unrecognized Eocene metamorphic core complex [cf. Seyfert, 1984].

## SUMMARY AND CONCLUSIONS

We obtained reliable paleomagnetic directions from 19 sites collected from middle Eocene dikes in the Bitterroot dome and Skalkaho slab. Their declinations differ from the expected direction, whereas their inclinations do not. The Bitterroot dome shows 3° of counterclockwise rotation at the 95% confidence level. The Skalkaho slab shows 26° of clockwise rotation at the 95% confidence level. Unfortunately, our sites in the Skalkaho slab may not completely average paleosecular variation. Future efforts will test this and determine whether the Skalkaho slab behaves as one independent block.

The paleomagnetic data and available geochronology [Bickford et al., 1981; Chase et al., 1983; Garmezy, 1983] show that the Bitterroot metamorphic core complex formed during regional Eocene crustal extension. The Bitterroot dome was pulled from under the Skalkaho slab and transported at least 25 km to the northwest. During translation, the dome rotated counterclockwise. At the southern end of the core complex, crustal extension stepped in echelon style to the southeast into the Big Hole Basin. The Skalkaho slab, bounded by these detachments and pinned at its northern margin, rotated clockwise. The regional geochronologic dates leave little doubt that this system of Eocene extension stretches from British Columbia via the Priest River complex, Boehls Butte, and the Bitterroot core complex to Tertiary basins in southwestern Montana and central Idaho.

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